$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = g(x)$$

linear aise ki differential coefficient ki power is 1

$a_0, a_1 \cdots a_n$ are Constant

Linear Differential Equations with Constant Coefficients

The general form is $f(D)y = X$, where $f(D)$ is a polynomial in the differential operator $D = \frac{d}{dx}$, and X is a function of x. The complete solution is given by:

$$y = \text{C.F.} + \text{P.I.}$$

I. Complementary Function (C.F.) - Solution to $f(D)y = 0$

The C.F. is found by solving the **Auxiliary Equation (A.E.)**: $f(m) = 0$, where D is replaced by m. The form of the C.F. depends on the nature of the roots $m_1, m_2, \ldots$ of the A.E.

1. **Case 1: Real and Distinct Roots** If the roots are $m_1, m_2, m_3, \ldots$ (all different), the C.F. is a linear combination of exponential terms.

$$\text{C.F.} = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + \ldots$$

Replace y by 1 and D by m, get value of $m_1, m_2, m_3 \cdots m_n$

2. **Case 2: Real and Repeated Roots** If a root m_1 is repeated k times (i.e., $m_1 = m_2 = \cdots = m_k$), we multiply the exponential term by a polynomial of degree $k-1$ in x.

auxiliary eqn = f(m)

$$\text{C.F.}_{\text{repeated}} = (C_1 + C_2 x + C_3 x^2 + \cdots + C_k x^{k-1}) e^{m_1 x}$$

Example: If $m = 2$ is repeated twice, C.F. $= (C_1 + C_2 x)e^{2x}$.

3. **Case 3: Complex Conjugate Roots** If the roots are a complex conjugate pair, $m_{1,2} = \alpha \pm i\beta$ (where $\beta \neq 0$), the C.F. is written in terms of sines and cosines, using Euler's formula.

$$\text{C.F.} = e^{\alpha x}(C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

Example: If $m = 3 \pm i\sqrt{5}$, then C.F. $= e^{3x}(C_1 \cos(\sqrt{5}x) + C_2 \sin(\sqrt{5}x))$.

4. **Case 4: Repeated Complex Conjugate Roots** If the complex pair $m_{1,2} = \alpha \pm i\beta$ is repeated k times (e.g., twice), the structure is similar to Case 2, with polynomial factors applied to the trigonometric terms.

$$\text{C.F.} = e^{\alpha x}[(C_1 + C_2 x)\cos(\beta x) + (C_3 + C_4 x)\sin(\beta x)]$$

Example: If $m = 1 \pm 2i$ is repeated twice, C.F. $= e^{x}[(C_1 + C_2 x)\cos(2x) + (C_3 + C_4 x)\sin(2x)]$.

II. Particular Integral (P.I.) - Rules for P.I. $= \frac{1}{f(D)}X$

A. Linearity and Fundamental Rules

- **Superposition:** $\frac{1}{f(D)}[X_1 + X_2] = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2$
- **Constant Multiple:** A constant k always factors out. $\frac{1}{f(D)}[k \cdot X] = k \cdot \left[\frac{1}{f(D)}X\right]$

B. Rules for Specific Functions

1. **Exponentials ($X = ke^{ax}$):**

$\sinh x = \frac{e^x - e^{-x}}{2}$

$$\text{PI} = k\frac{1}{f(D)}e^{ax} = k\frac{e^{ax}}{f(a)} \quad \text{if } f(a) \neq 0$$

$\cosh x = \frac{e^x + e^{-x}}{2}$

If $f(a) = 0$ (Case of Failure), multiply by x and differentiate the denominator $f(D)$:

$$\text{PI} = k \cdot x\frac{1}{f'(D)}e^{ax} = k\frac{xe^{ax}}{f'(a)}$$

a^x आ गया तो $= e^{\ln a^x}$

यहाँ trigo के manipulation भी करवा सकते हैं, जैसे $X = \sin 5x \cos 2x$ ऐसे - Convert to $\sin x$ and $\cos x$

2. **Trigonometric** ($X = k\sin(ax+b)$ or $k\cos(ax+b)$): Substitute $D^2 = -a^2$ into $f(D)$. The phase angle ($+b$) is ignored in the substitution.

$$\text{PI} = k\frac{1}{f(D)}\sin(ax+b) = k\frac{1}{\phi(D^2)}\sin(ax+b) \xrightarrow{D^2\to -a^2} k\frac{\sin(ax+b)}{\phi(-a^2)} \quad \text{if } \phi(-a^2) \neq 0$$

जैसे $\cos^2 x$ आ गया तो convert it into $\cos 2x$

$\cos^2 x = \dfrac{1+\cos 2x}{2}$

If a linear D term remains, multiply by the conjugate to create D^2 in the denominator:

$$\frac{1}{D-k}\cos(ax+b) = \frac{D+k}{D^2-k^2}\cos(ax+b)$$

D^3 के ques में उसे $D^2(D)$ करो फिर linear आये तो उसके साथ conjugate conjugate रखेंगे।

If $f(D) = D^2 + a^2$ (Case of Failure):

$$\frac{1}{D^2+a^2}\cos(ax) = \frac{x\sin(ax)}{2a}$$

3. **Exponential Shift Rule** ($X = e^{ax}V(x)$): This rule is universally valid for any $V(x)$.

$$\text{PI} = \frac{1}{f(D)}[e^{ax}V(x)] = e^{ax}\left[\frac{1}{f(D+a)}V(x)\right]$$

अब आप $\dfrac{V(x)}{f(D+a)}$ का PI निकाले।

4. **General Method** ($\tan(x)$, $\sec(x)$, etc.): For functions without a simple operator rule, solve by factoring and integrating.

$$\text{If } \frac{1}{f(D)} = \frac{1}{D-r} \text{ (after partial fractions), then: } \frac{1}{D-r}X(x) = e^{rx}\int e^{-rx}X(x)\,dx$$

* Yaha par f(D) को factor के form me Convert करके partial fraction Bana ke formula लगाना है।

5. x^m, $m \in \mathbb{N}$ (Yaha linear $(3x-2)$ या polynomial भी हो सकता है।)

$\text{PI} = \dfrac{x^m}{f(D)}$, यहाँ $f(D)$ को $1+f(D)$ या $1-f(D)$ में Convert करो By Taking Constant term Common

$\text{PI} = \dfrac{x^m}{1+f(D)} = (1+f(D))^{-1}x^m$ → ऐसा बन जायेगा।

Now Use these formulas:-

$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$

$(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots$

* $(1\pm f(D))^{-1}$ को vahi tak expand करना जितनी x की degree हो like x^2 तो x^2 तक उसको expand

then Multiply x^m inside and Take derivative

Scanned with OKEN Scanner

⊙ Homogenous linear Differential equations : Euler - Cauchy Equation

General form $a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = g(x)$

where a_i, $0 \leq i \leq n$ are Constant

① Let $x = e^z$ and $z = \ln x$

② Replace $x \frac{dy}{dx} = Dy$ $\quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$ $\quad D \equiv \frac{d}{dz}$

$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y$ and so on

③ After This replacement you get a linear DE with constant coefficients, in variable z

* variable z में आयेगी। उसे x में Convert करो!!

CF aur PI will be $f(z)$, Convert to x by putting $z = \ln x$

⊙ Legendre's linear DE

General form $a_0 (ax+b)^n \frac{d^n y}{dx^n} + a_1 (ax+b)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 (ax+b)^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \cdots a_n y = f(x)$

put $(ax+b) = e^z$, $x = \left(\frac{e^z - b}{a}\right)$ $\quad z = \ln(ax+b)$

① Here replace $K(ax+b)^2 \frac{d^2 y}{dx^2} = K D(D-1) \times (\text{derivative of } ax+b)^2$

$= K a^2 D(D-1)$

$(ax+b)^3 \frac{d^3 y}{dx^3} = a^3 D(D-1)(D-2)$

* Now it will get converted to LDE with constant coefficients

⊙ Method of variation of parameters : A Method to get PI of a differential equation

* Method is general but is restricted to order 2 only, generally.

① Step 1 : get the CF of DE, say $CF = C_1 y_1 + C_2 y_2$ (कुछ भी y_1, y_2)

② Step 2 : get y_1 and y_2

③ Step 3 : get Wronskian (W), $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

④ Now $u = -\int \frac{y_2 X}{W}$ $\quad v = \int \frac{y_1 X}{W}$

⑤ Now $PI = u y_1 + v y_2$

...od of variation of parameters

...1: Calculate CF

$CF = C_1 y_1 + C_2 y_2$ $\qquad$ C_1 के साथ y_1 और C_2 के साथ y_2

you need to have y_1 and y_2

Step 2 : Calculate value of W

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Step 3 :

$$u = -\int \frac{y_2 X\,dx}{W} \qquad V = \int \frac{y_1 X\,dx}{W}$$

Step 4 : Now the Particular Integral $PI = uy_1 + Vy_2$

and solution = CF + PI

Que) $\frac{d^2y}{dx^2} + y = \sec x \tan x$

$(D^2+1) = 0 \Rightarrow D = 0 \pm i \qquad \alpha = 0 \quad \beta = 1$

$CF = C_1 \cos x + C_2 \sin x$

$y_1 = \cos x \qquad y_2 = \sin x \qquad X = \sec x \tan x = \sec x$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1. \qquad W = 1$$

$$u = -\int \sin x \times \sec x \tan x\,dx = -\int \tan^2 x\,dx$$

WKT $\quad \cos 2\theta = \frac{1-\tan^2 x}{1+\tan^2 x}$

$$\frac{1+\tan^2 x}{1-\tan^2 x} = \frac{1}{\cos 2x}$$

applying Componendo and dividendo

$$\frac{1}{\tan^2 x} = \frac{1+\cos 2x}{1-\cos 2x}$$

$$\tan^2 x = \frac{1-\cos 2x}{1+\cos 2x}$$

$\cos 2x = 2\cos^2 x - 1$

$1+\cos 2x = 2\cos^2 x$

$$-\int \left(\frac{1}{1+\cos 2x} - \frac{\cos 2x}{1+\cos 2x} \right) dx$$

$$-\int \left(\frac{1}{2\cos^2 x} - \left(\frac{1+\cos 2x}{1+\cos 2x} - \frac{1}{1+\cos 2x} \right) \right) dx$$

$$-\int \left(\frac{2}{1+\cos 2x} - 1 \right) dx$$

$$-\int (\sec^2 x - 1)\,dx = -(\tan x - x) = x - \tan x$$

$V = \int \cos x \cdot \sec x \tan x \, dx = \int \tan x \, dx = \ln \sec x$ ILATE

$\int (1) \tan x \, dx = I \int \tan x -$ $\int \frac{\sin x}{\cos x} dx$

$V = \ln \sec x$ $\quad u = x - \tan x$

$PI = (x - \tan x)\cos x + (\ln \sec x)\sin x$

$y = (C_1 \cos x + C_2 \sin x) + (x - \tan x)\cos x + (\ln \sec x)\sin x$

Que) $(D^2 + 3D + 2)y = e^{e^x}$

$D^2 + 2D + D + 2 = 0$

$D(D+2) + 1(D+2) = 0$

$(D+1)(D+2)$

$D = -2 \quad D = -1$

$CF = C_1 e^{-x} + C_2 e^{-2x}$ descending is imp*

$y_1 = e^{-x} \quad y_2 = e^{-2x}$

$$W = \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -2e^{-3x} + e^{-3x} = -e^{-3x}$$

$\boxed{W = -e^{-3x}}$

$u = -\int \frac{y_2 X}{W} dx = +\int \frac{e^{-2x} \cdot e^{e^x}}{+e^{-3x}} dx$

$= \int e^{e^x - 2x + 3x} dx = \int e^{(e^x + x)} dx$

$= \int e^{e^x} \cdot e^x \, dx$ $\quad e^x = t$

$e^x \frac{dx}{dt} = 1 \quad e^x dx = dt$

$u = \int e^t dt = e^{e^x}$

$u = e^{e^x}$

ILATE

$V = \int \frac{y_1 X}{W} dx = \frac{e^{-x} \cdot e^{e^x}}{e^{-3x}} = \int e^{e^x - x + 3x} = \int e^{e^x} \cdot e^{2x} dx$ $\quad$ I: t, II: e^t; 1, e^t; 0, e^t

$= \int e^{e^x} \cdot e^x \cdot e^x \, dx = \int e^t t \, dt = te^t - e^t$

$V = e^x e^{e^x} - e^{e^x} = e^{(e^x + x)} - e^{e^x} = e^{e^x}(e^x - 1)$

$V = e^{e^x}(e^x - 1)$

$PI = uy_1 + vy_2 = (e^{e^x} e^{-x}) + e^{e^x}(e^x - 1)e^{-3x}$

$= e^{(e^x - x)} + e^{(e^x - 3x)}(e^x - 1)$

$GS = y = C_1 e^{-x} + C_2 e^{-2x} + e^{(e^x - x)} + e^{e^x - 3x}(e^x - 1)$